

Q1.

The curve, C , has equation $y = \frac{x}{\cosh x}$

- (a) Show that the x -coordinates of any stationary points of C satisfy the equation $\tanh x = \frac{1}{x}$

$$\tanh x = \frac{1}{x}$$

(3)

- (b) (i) Sketch the graphs of $y = \tanh x$ and $y = \frac{1}{x}$ on the axes below.

$$y = \frac{x}{\cosh x}$$

(2)

- (ii) Hence determine the number of stationary points of the curve C .

(1)

- (c) Show that $\frac{d^2y}{dx^2} + y = 0$ at each of the stationary points of the curve C .

(4)

(Total 10 marks)

Q2.

- (a) Using the definition $\tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$, show that, for $|x| < 1$,

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

(3)

- (b) Hence, or otherwise, show that $\frac{d}{dx} (\tanh^{-1} x) = \frac{1}{1-x^2}$.

(3)

- (c) Use integration by parts to show that

$$\int_0^{\frac{1}{2}} 4 \tanh^{-1} x \, dx = \ln \left(\frac{3^m}{2^n} \right)$$

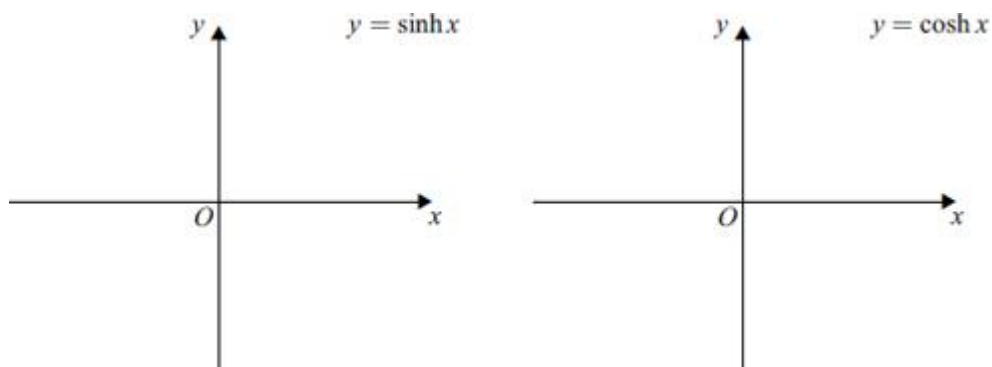
where m and n are positive integers.

(5)

(Total 11 marks)

Q3.

- (a) (i) Sketch on the axes below the graphs of $y = \sinh x$ and $y = \cosh x$.



(3)

- (ii) Use your graphs to explain why the equation

$$(k + \sinh x) \cosh x = 0$$

where k is a constant, has exactly one solution.

(1)

- (b) A curve C has equation $y = 6 \sinh x + \cosh^2 x$. Show that C has only one stationary point and show that its y -coordinate is an integer.

(5)

(Total 9 marks)

Q4.

- (a) (i) Show that

$$\frac{d}{du} (2u\sqrt{1+4u^2} + \sinh^{-1} 2u) = k\sqrt{1+4u^2}$$

where k is an integer.

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(4)

- (ii) Hence show that

$$\int_0^1 \sqrt{1+4u^2} \, du = p\sqrt{5} + q \sinh^{-1} 2$$

where p and q are rational numbers.

(2)

- (b) The arc of the curve with equation $y = \frac{1}{2} \cos 4x$ between the points where $x = 0$ and $x = \frac{\pi}{8}$ is rotated through 2π radians about the x -axis.

- (i) Show that the area S of the curved surface formed is given by

$$S = \pi \int_0^{\frac{\pi}{8}} \cos 4x \sqrt{1 + 4 \sin^2 4x} \, dx$$

(2)

- (ii) Use the substitution $u = \sin 4x$ to find the exact value of S .

(4)

(Total 12 marks)

Q5.

A curve has cartesian equation

$$y = \frac{1}{2} \ln(\tanh x)$$

- (a) Show that

$$\frac{dy}{dx} = \frac{1}{\sinh 2x}$$

(4)

- (b) The points A and B on the curve have x -coordinates $\ln 2$ and $\ln 4$ respectively. Find the arc length AB , giving your answer in the form $p \ln q$, where p and q are rational numbers.

(8)

(Total 12 marks)

Q6.

- (a) Show that

$$\frac{1}{4}(\cosh 4x + 2 \cosh 2x + 1) = \cosh^2 x \cosh 2x$$

(3)

- (b) Show that, if $y = \cosh^2 x$, then

$$1 + \left(\frac{dy}{dx}\right)^2 = \cosh^2 2x$$

(3)

- (c) The arc of the curve $y = \cosh^2 x$ between the points where $x = 0$ and $x = \ln 2$ is rotated through 2π radians about the x -axis. Show that the area S of the curved surface formed is given by

$$S = \frac{\pi}{256} (a \ln 2 + b)$$

where a and b are integers.

(7)

(Total 13 marks)

Q7.

- (a) Prove that the curve

$$y = 12 \cosh x - 8 \sinh x - x$$

has exactly one stationary point.

(7)

- (b) Given that the coordinates of this stationary point are (a, b) , show that $a + b = 9$.

(4)

(Total 11 marks)

Q8.

- (a) Using the identities

$$\cosh^2 t - \sinh^2 t = 1, \tanh t = \frac{\sinh t}{\cosh t} \text{ and } \operatorname{sech} t = \frac{1}{\cosh t}$$

show that:

(i) $\tanh^2 t + \operatorname{sech}^2 t = 1;$

(2)

(ii) $\frac{d}{dt} (\tanh t) = \operatorname{sech}^2 t;$

(3)

(iii) $\frac{d}{dt} (\operatorname{sech} t) = -\operatorname{sech} t \tanh t.$

(3)

- (b) A curve C is given parametrically by

$$x = \operatorname{sech} t, y = 4 - \tanh t$$

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- (i) Show that the arc length, s , of C between the points where $t = 0$ and $t = \frac{1}{2} \ln 3$ is given by

$$s = \int_0^{\frac{1}{2} \ln 3} \operatorname{sech} t \, dt$$

(4)

- (ii) Using the substitution $u = e^t$, find the exact value of s .

(6)

(Total 18 marks)

Q9.

- (a) Given that $u = \cosh^2 x$, show that $\frac{du}{dx} = \sinh 2x$.

(2)

- (b) Hence show that

$$\int_0^1 \frac{\sinh 2x}{1 + \cosh^4 x} dx = \tan^{-1}(\cosh^2 1) - \frac{\pi}{4}$$

(5)

(Total 7 marks)

Q10.

- (a) Show that

$$\frac{d}{dx} \left(\cosh^{-1} \frac{1}{x} \right) = \frac{-1}{x\sqrt{1-x^2}}$$

(3)

- (b) A curve has equation

$$y = \sqrt{1-x^2} - \cosh^{-1} \frac{1}{x} \quad (0 < x < 1)$$

Show that:

(i) $\frac{dy}{dx} = \frac{\sqrt{1-x^2}}{x};$

(4)

- (ii) the length of the arc of the curve from the point where $x = \frac{1}{4}$ to the point where $x = \frac{3}{4}$ is $\ln 3$.

(5)

(Total 12 marks)

Q11.

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- (a) Given that $y = \ln \tanh \frac{x}{2}$, where $x > 0$, show that

$$\frac{dy}{dx} = \operatorname{cosech} x$$

(6)

- (b) A curve has equation $y = \ln \tanh \frac{x}{2}$, where $x > 0$. The length of the arc of the curve between the points where $x = 1$ and $x = 2$ is denoted by s .

- (i) Show that

$$s = \int_1^2 \coth x dx$$

(2)

- (ii) Hence show that $s = \ln(2 \cosh 1)$.

(4)

(Total 12 marks)

Q12.

- (a) Given that $y = \operatorname{sech} t$, show that:

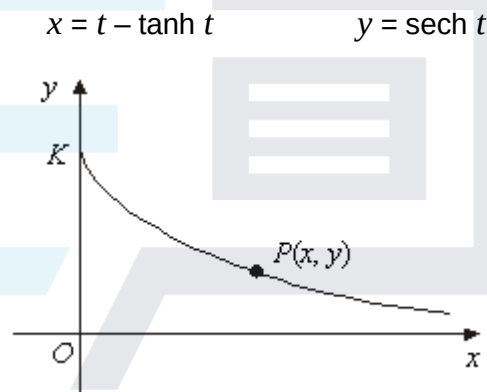
(i) $\frac{dy}{dt} = -\operatorname{sech} t \tanh t$

(3)

(ii) $\left(\frac{dy}{dt}\right)^2 = \operatorname{sech}^2 t - \operatorname{sech}^4 t$.

(2)

- (b) The diagram shows a sketch of part of the curve given parametrically by



The curve meets the y -axis at the point K , and $P(x, y)$ is a general point on the curve. The arc length KP is denoted by s . Show that:

(i) $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \tanh^2 t$,

(4)

(ii) $s = \ln \cosh t$;

(3)

(iii) $y = e^{-s}$.

(2)

- (c) The arc KP is rotated through 2π radians about the x -axis. Show that the surface area generated is $2\pi(1 - e^{-s})$

(4)

(Total 18 marks)

Q13.

- (a) Use the definitions

$$\sinh \theta = \frac{1}{2}(e^{\theta} - e^{-\theta}) \quad \text{and} \quad \cosh \theta = \frac{1}{2}(e^{\theta} + e^{-\theta})$$

to show that:

(i) $2 \sinh \theta \cosh \theta = \sinh 2\theta;$

(2)

(ii) $\cosh^2 \theta + \sinh^2 \theta = \cosh 2\theta.$

(3)

(b) A curve is given parametrically by

$$x = \cosh^3 \theta, \quad y = \sinh^3 \theta$$

(i) Show that

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \frac{9}{4} \sinh^2 2\theta \cosh 2\theta$$

(6)

(ii) Show that the length of the arc of the curve from the point where $\theta = 0$ to the point where $\theta = 1$ is

$$\frac{1}{2} \left[(\cosh 2)^{\frac{3}{2}} - 1 \right]$$

(6)

(Total 17 marks)

Q14.

The curve C has equation

$$y = \cosh x - 3 \sinh x$$

(a) (i) The line $y = -1$ meets C at the point $(k, -1)$.

Show that

$$e^{2k} - e^k - 2 = 0$$

(3)

(ii) Hence find k , giving your answer in the form $1 \ln a$.

(4)

(b) (i) Find the x -coordinate of the point where the curve C intersects the x -axis, giving your answer in the form $p \ln a$.

(4)

(ii) Show that C has no stationary points.

(3)

- (iii) Show that there is exactly one point on C for which $\frac{d^2 y}{dx^2} = 0$.

(1)

(Total 15 marks)



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